HACKATHON 7.0

NETRA IDENTITY

AI-Powered Offline Facial Recognition & Anti-Spoofing Attendance System for NHAI Field Personnel

42ms	100%	6.3MB	7-Layer
Pipeline Latency	Offline Capable	ML Models	Security

THE CHALLENGE

The Problem with Field Attendance

NHAI Field Reality

National Highways Authority of India deploys thousands of field personnel across remote highway locations — many with zero or unreliable internet connectivity.

Existing Solutions Cannot:

- › Authenticate without internet
- › Detect photo/video spoof attacks
- › Encrypt data locally at rest
- › Auto-sync when connectivity resumes

✗ Cloud Systems Fail Offline

Traditional biometric & cloud-based facial recognition require persistent internet. No connectivity = no attendance.

✗ Proxy Attendance / Buddy Punching

Manual log books and basic biometrics are easily spoofed — printed photos, screen replays, and proxy sign-ins go undetected.

✗ No Secure Local Fallback

Even when internet is available, data is stored insecurely with no encryption or tamper detection on the device.

OUR SOLUTION

Introducing Netra Identity

"A fully offline-capable, AI-powered facial recognition and anti-spoofing attendance system — military-grade security that fits in your pocket."

On-Device AI

3-model TFLite pipeline runs 100% locally. No cloud required for authentication.

Anti-Spoofing

Passive + active liveness detection blocks photos, screens, and replay attacks.

AES-256 Encrypted

SQLCipher encrypts all biometric data at rest. Keys stored in OS Keychain.

Auto Cloud Sync

Background sync to AWS S3 on connectivity restore with exponential backoff.

Built on React Native (Expo) · TFLite via JSI Bridge · AWS Serverless Backend · Cross-Platform iOS + Android

ARCHITECTURE

System Architecture

MOBILE DEVICE (Edge)

ML PIPELINE

BlazeFace
229KB · Face Detection

MobileFaceNet
5.94MB · Embeddings

MiniFASNet
156KB · Anti-Spoof

SERVICES

StorageService
SQLCipher AES-256

SecurityService
PBKDF2 · SHA-256 · Keychain

SyncService
NetInfo · Exponential Backoff

Auto-sync ▶

◀ Pull enrollments

AWS CLOUD (us-east-1)

API Gateway
HTTPS + CORS

Lambda (Node 18)
Serverless Sync Handler

S3 Bucket
SSE-AES-256 Data Lake

USER → Enrollment (5 angles) → MobileFaceNet Embeddings → SQLCipher DB → Liveness Challenge → Match (cosine ≥0.6) → Attendance Saved → AWS S3

AI / ML PIPELINE

3-Stage On-Device ML Pipeline

01 BlazeFace

Face Detection

229 KB	128×128	5-10ms
Model Size	Input	Latency

Scout model scans frames at >15fps. 896 classification anchors with custom sigmoid activation. Detects face bounding box for downstream crops.

02 MobileFaceNet

Face Embedding

5.94 MB	112×112	15-25ms
Model Size	Input	Latency

Lightweight CNN generates a 128-dimensional embedding vector per face. Normalized via (pixel/127.5)–1. Stored with SHA-256 checksum integrity.

03 MiniFASNet

Anti-Spoofing

156 KB	80×80	5-10ms
Model Size	Input	Latency

Trained on CelebA-Spoof dataset. ImageNet normalization. 3-class softmax (real vs. 2 spoof types). Score index[1] = real face probability.

FACIAL RECOGNITION

128-Dim Embedding & Cosine Matching

ENROLLMENT — 5-Angle Capture

FR	LE	RI	UP	DO
FRONT	LEFT	RIGHT	UP	DOWN

1. Vision Camera captures JPEG at 1280×720
2. Skia GPU: center crop → resize to 112×112
3. Normalize: (pixel / 127.5) − 1.0 → range [−1, 1]
4. MobileFaceNet inference → 128-dim float vector
5. Validate (no NaN / all-zeros) + SHA-256 checksum
6. Store in SQLCipher DB · 5 embeddings per person

AUTHENTICATION — Cosine Matching

```
similarity = (A · B) / (|A| × |B|)
```

· Live camera generates embedding every ~1 second
· Compare against ALL enrolled embeddings in local DB
· Best-match algorithm: highest similarity across all angles
· Threshold: Cosine Similarity ≥ 0.6 → Identity Confirmed
· Scale-invariant: works regardless of embedding magnitude
· No network needed — all matching happens on-device

128-dim	≥ 0.6	5 angles	~1s
Embedding Vector	Match Threshold	Per Person Enrolled	Per-Frame Refresh

ANTI-SPOOFING

Passive + Active Liveness Detection

🔬 PASSIVE LIVENESS — MiniFASNet

Trained on CelebA-Spoof dataset. Detects texture anomalies in printed photos and screen displays.

Normalization:	RGB ImageNet (not [-1,1])
Mean (R,G,B):	0.485 · 0.456 · 0.406
Std (R,G,B):	0.229 · 0.224 · 0.225
Output Classes:	3 ($spoof_1$, real, $spoof_2$)
Real Score:	softmax[index 1] × 100
Threshold:	Average ≥ 60% over 5 seconds

👁 ACTIVE LIVENESS — Blink Detection

5-second challenge prompts user to blink. Continuous MiniFASNet scoring samples every frame.

- Challenge Duration: 5 seconds
- Sampling: Continuous per-frame scoring
- Decision: Average score across all frames
- Blink prompt shown to user at ~2s mark
- Failure → Spoof Alert + Re-attempt
- Success → Attendance logged to SQLCipher

Threats Blocked:

- ✗ Printed Photo
- ✗ Screen Replay
- ✗ Video Loop
- ✗ Static Mask
- ✗ Deepfake Injection

OFFLINE + SYNC

Offline-First Architecture & Auto Cloud Sync

SQLCipher Database Schema

Personnel

id · name · role · embeddings (128-dim×5) · sha256_checksum

AttendanceLog

id · personnel_id (FK) · timestamp · result · synced flag

SyncQueue

id · record_type · payload_json · retry_count · created_at

SyncService — Auto Cloud Sync

- NetInfo listener detects connectivity restore
- Batch upload: 20 records per HTTP request
- Exponential backoff: 1s → 2s → 4s on failure
- Serial timestamps prevent data overwrites
- Pulls remote enrollments → bi-directional sync
- Auto-purge synced records older than 24 hours

AWS: API Gateway → Lambda (Node 18) → S3 (SSE-S3 AES-256) · us-east-1

AES-256 SQLCipher + PBKDF2 Key Derivation + SHA-256 Embedding Integrity

Works 100% offline forever

Auto-sync on connectivity restore

Bi-directional device sync

Zero plaintext data stored

SECURITY

7-Layer Security Stack

L1
Device Integrity
JailMonkey root/jailbreak detection · mock GPS injection check

L2
Key Derivation
PBKDF2 (1,000 iterations) from UUID + app secret → encryption key

L3
Secure Key Storage
Android Keystore / iOS Keychain for encryption key persistence

L4
Database Encryption
SQLCipher AES-256 encrypts entire database file at rest

L5
Embedding Integrity
SHA-256 checksum on each embedding; tampered data rejected on read

L6
Transport Security
HTTPS (TLS 1.2+) via AWS API Gateway for all sync operations

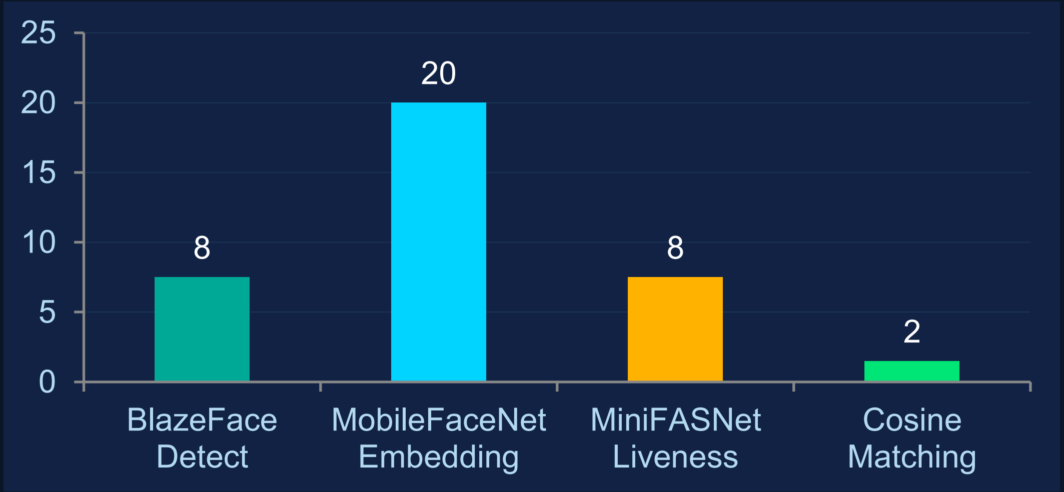
L7
Cloud Encryption
S3 SSE-S3 AES-256 server-side encryption for stored records

Zero trust model: Device → Key → Crypto → DB → Embeddings → Network → Cloud — every layer independently verified and encrypted

PERFORMANCE

Performance & Optimization

42ms	45MB	6.3MB	100%
Pipeline Latency	RAM Usage	Total ML Models	Offline Capable



Key Optimizations

- › JSI C++ Bridge — zero JS serialization
- › Single model load, shared via Context
- › 400ms frame throttle + mutex lock
- › arm64-v8a split ABI (−60% native libs)
- › Hermes pre-compiled JS bytecode
- › ProGuard R8 dead code elimination
- › Skia GPU-accelerated image processing

TECHNOLOGY STACK

Full Technology Stack

Frontend

- React Native 0.81.5
- Expo ~54.0.0
- TypeScript ^5.9.3
- React Navigation v6
- Vision Camera 4.7.3
- React Native Skia 2.2.12

AI / ML

- react-native-fast-tflite (JSI)
- BlazeFace · 229 KB
- MobileFaceNet · 5.94 MB
- MiniFASNet · 156 KB
- vision-camera-resize-plugin

Backend & Cloud

- AWS Lambda Node.js 18
- Amazon API Gateway
- Amazon S3 (us-east-1)
- Serverless Framework IaC
- Axios HTTP client

Security & Storage

- SQLCipher AES-256
- react-native-keychain
- CryptoJS PBKDF2 + SHA-256
- JailMonkey (root detection)
- UUID v4 identifiers

Build: Hermes Engine · ProGuard R8 · Split ABI (arm64-v8a) · PNG Crunching · React Native NetInfo · Serverless IaC on AWS

INNOVATION

Engineering Challenges Solved

TFLite on React Native

react-native-fast-tflite JSI bridge — C++ native inference, zero bridge serialization overhead

GPU Image Preprocessing

React Native Skia for GPU-accelerated center crop, resize, and pixel extraction without native modules

BlazeFace Output Parsing

Custom sigmoid on 896 classification anchors; manual parsing of regressor (14,336) + classifier tensors

MiniFASNet Normalization

Empirically discovered RGB ImageNet normalization yields correct results — not the [-1,1] range used by face models

Offline-First + Sync

3-table schema with SyncQueue + FK constraints; bi-directional sync with serial timestamps preventing overwrites

Database Encryption in RN

SQLCipher via react-native-sqlite-storage with PBKDF2-derived key; graceful fallback for non-Keychain environments

ROADMAP

Future Scope & Roadmap

v1.1

📍 GPS Location Tagging

Attach geo-coordinates to each attendance record for highway site verification and compliance.

v1.2

🔒 SSL Certificate Pinning

Pin API Gateway certificate to prevent MITM proxy attacks.

v1.3

📊 Admin Dashboard

Web-based dashboard for HR managers to view and analyze synced attendance records.

v2.0

🧬 3D Depth Liveness

Structured light / ToF sensor depth data for physical spoof resistance.

v2.1

🔄 OTA Model Updates

Over-the-air TFLite model updates via S3 — improve accuracy without app store releases.

v2.2

📡 Bluetooth P2P Sync

Inter-device sync via Bluetooth/NFC — share enrollments even with no internet.

CONCLUSION

NETRA IDENTITY

Military-Grade Identity Verification. Pocket-Sized.

- ✓ 3-model on-device ML pipeline — 100% offline authentication
- ✓ Passive + active liveness detection — photo & replay attacks blocked
- ✓ 128-dim MobileFaceNet embeddings — cosine similarity matching
- ✓ AES-256 SQLCipher + PBKDF2 + SHA-256 — 7-layer security
- ✓ 42ms pipeline · 45MB RAM · 6.3MB models — ultra-lightweight
- ✓ Auto AWS sync + bi-directional enrollment — serverless scalable backend

"Netra Identity proves that military-grade identity verification doesn't require a data center — it fits in your pocket."